# Ricardo Rodrigues da Silveira Filho

Biologist (PhD)
Data Scientist
Developer

## CONTACT

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## SKILLS

**Languages**
TypeScript, Python, R, SQL (PL/pgSQL), HTML, CSS, Shell

**Backend**
Node.js, FastAPI, Prisma, PostgreSQL, Supabase, Redis, PowerSync

**Frontend and mobile**
React, Next.js, Tailwind, PWA, Capacitor (Android), Shiny

**Infrastructure and testing**
Git, Docker, nginx, GitHub Actions, Cloudflare, Vitest, testthat

**Data and analytics**
ecological statistics, geospatial data (QGIS, PostGIS), Power BI (DAX, Power Query), ggplot2, Quarto and R Markdown, emissions inventory (GHG Protocol)

**Applied AI**
agent-assisted development (Claude Code), chatbots (Blip), team AI adoption

**Product and management**
Product Owner, requirements gathering, ADRs, Scrum, project management (PMI), pre-sales and R&D grant proposals

## LANGUAGES

Portuguese (native)
English (fluent)
Spanish (basic)

## Summary

Data scientist and software developer with a PhD in Zoology, two postdocs and more than fifteen years in environmental consulting. I know the data from the moment it's collected, and I build the system that receives it. For the last two years I led the Innovation area of an environmental consultancy as Product Owner and Tech Lead, with internal systems used across the company, an ESG management SaaS platform, a field data collection suite and BI dashboards for clients. Before that, I built and ran a customer service chatbot with thousands of monthly users and helped create a delivery app for neighbourhood shops. I code in TypeScript, Python and R, lead teams with clear scope and documented decisions, and use generative AI in development, with the gain measured in hours.

## Experience

**Seteg** 2024–present
*Innovation Analyst II* (Jun 2025 – present) · *Environmental Analyst* (Nov 2024 – Jun 2025) · environmental and geological consultancy

- As Product Owner, conceived and led the project and time management system that replaced a paid tool and is used by the **whole company (about 45 people)**, with a closed beta as the go-live gate.
- Led the consolidation of **14 internal systems into 2**, including 8 request systems used by **about 40 people in 5 departments**: requirements gathering, delivery acceptance, migration order and permission matrix.
- Designed the data model of a field data collection suite (PostgreSQL/PostGIS, Supabase with RLS, offline-first sync) and led the company's first project management system (Fastify, Prisma, React), with ERP integration and 189 tests, reviewing the code and coding alongside the team when needed.
- Developed the goals module of an ESG management SaaS platform funded by an innovation grant, and drove the final deliveries through to production deployment.
- Coordinated a team of 4 (3 developers and 1 designer), with direct management and an individual development plan for one developer; Innovation's point of contact with the board and the PMO.
- Wrote the area's **operating manual**: how requirements are written, the git workflow, how project boards are kept and the delivery cycle, from request to handover.
- Measured the productivity gain from generative AI in three departments (**about 70% less time**, based on logged hours) and wrote the company's first AI usage policy.
- Delivered **GHG Protocol** emissions inventories, in the ESG platform and in emissions and target dashboards (Power BI, DAX), and a hydropower generation analytics solution for an energy sector client (FastAPI, trend tests).
- Built tools for field work and management: a **field photo metadata extractor** (EXIF, OCR and reverse geocoding, shipped as a single executable), a forest inventory progress dashboard for a client (a static site reading the field spreadsheet on SharePoint through Microsoft Graph) and a weekly team hours dashboard.

**Conciso TI** 2020–2024
*Data Scientist* (Feb 2020 – Oct 2024) · technology

- Conceived, built and ran **Nubus**, the company's customer service chatbot on the Blip platform: **3,000 to 6,000 unique users a month** (peak of 10,000) and a rating always above 4.5 out of 5 (usually **4.7 to 4.8**). I was the point of contact with Blip and automated the **monthly usage reports** in R (ggplot2, R Markdown).
- Built the **monthly demand forecast** for customer service: time series modelling with **Prophet in R** to estimate how many people would reach the chatbot the following month. Customer service, sales and the board used it to prepare for the month ahead.
- Co-conceived **Na Esquina** with the two other developers on the team: a delivery app for small and mid-sized grocery stores, built in six months during the pandemic, before the major delivery apps served neighbourhood shops. I created the name and the brand and built the Firebase database, with my own pipeline that listed products and pulled their images from the web to build the catalog.

**Elementus Soluções Ambientais** 2023–2025
*Senior Project Analyst* (Oct 2023 – Apr 2025) · environmental consultancy

- Technical lead (registered with the Biology Council) for sea turtle monitoring at the Port of Natal: quarterly boat surveys, georeferenced records and technical reports to meet the licence conditions.

**Academic research** 2010–2023

UFRN/PPBio · UNEMAT · UFPB

- *Postdoc, UFRN/PPBio* (2022–2023): pipeline for Brazilian vertebrate checklists and genetic data coverage (BOLD, GenBank), which grew into tetrapoDNA.
- **Collaborating researcher at the Knowledge Center for Biodiversity** (Brazilian biodiversity research network led by UFMG) since 2022, after my PPBio postdoc. Through the network: two letters in *Science* (2023 and 2024) and a paper in *Perspectives in Ecology and Conservation* (2026).
- *Postdoc, UNEMAT* (2018–2019): woody community dynamics in the Cerrado; co-supervision, thesis committees and R courses.
- *MSc and PhD in Zoology, UFPB* (2010–2017): mimicry and phylogeography of snakes, including a one-month research visit to the Natural History Museum, London (2011), surveying its coral snake collection with the museum's researchers.
- **Assessor for the Pernambuco state Red List of amphibians and reptiles** (2014 workshop, IUCN criteria, snakes group) and co-author of the state's official lists of threatened species (amphibians, 2015; reptiles, 2017).
- **15 peer-reviewed papers and 1 accepted (2026)**, including two letters in *Science* (2023 and 2024).

**Freelance environmental and statistical consultant** 2010–present

- Herpetofauna work for environmental licensing in energy, mining and infrastructure, hired by consultancies (ADL, Life, Maron, Ophian, Seteg, Spizaetus, Táxon): monitoring programmes and baseline surveys for impact assessments and licensing reports at wind and solar complexes (**ENEL, ENGIE, AES Brasil, Elera**), a bauxite mine in the Amazon (**Mineração Rio do Norte**), a railway (Transnordestina), a thermal power plant (CELSE) and a gas pipeline (TAG).
- Ecological analyses for herpetofauna monitoring in environmental licensing (richness, diversity, similarity, NMDS, rarefaction), with didactic reports in R and Quarto (2023–present).
- Data analysis for the **impact assessment reports on how the Manati gas pipeline and platform affect small-scale fisheries in Bahia, delivered to Petrobras** (1st to 4th report, 2023–present): catch, CPUE, cost and income statistics by fishery in the 1st report; for the 4th, a **20-year landing database (over 1 million records**: about 366,000 trips and 700,000 species records) rebuilt in R from raw spreadsheets and a legacy Python database, an audit and correction of the historical series, trend tests (Mann-Kendall, Sen) by fishing ground and distance to the project, and maps and tables generated by the pipeline.
- Co-founder of **Consulte um Cientista** (2020–2023), a statistics and research consultancy for companies and individuals, with analyses in R and didactic Quarto reports.

**Barbosa & Barbosa** 2008–2009

*CAD Drafter* (2008 – 2009)

- Technical drawings for architectural projects in AutoCAD.

## Projects

- **tetrapoDNA and traqueophyDNA.** Public dashboard showing which Brazilian tetrapods (about 4,800 amphibian, reptile, bird and mammal species) still have no DNA sequence in BOLD or GenBank. R pipeline with synonym checks (CTFB, GBIF) to avoid false gaps, weekly updates via GitHub Actions and a GitHub Pages site. *traqueophyDNA*, under construction, applies the same method to the roughly 35,000 native vascular plants in Flora e Funga do Brasil, with synonyms from WCVP and the NCBI taxonomy.
  rsilveirafh.github.io/tetrapoDNA
- **claude-plugins.** Open-source catalog of my Claude Code plugins. *banca-simulada* reviews research proposals, from undergraduate theses to faculty hiring, the way a committee would: it points to the passage, the severity and a suggestion across eight axes, and lists the likely questions. *memoria-skills* turns a Markdown vault into a memory system with human review.
  github.com/rsilveirafh/claude-plugins
- **R packages for species checklists (amphiBR and ReptiliaBR).** Open-source contributions: the 2026 Brazilian amphibian checklist (1,251 species) and threat status from the 2026 national red list.
- **Online stores and custom websites.** Two confectionery stores live since August 2026, built on the same template (Next.js, Supabase, PIX instant payments, PWA): product configurator, server-side pricing, a real-time order panel and 259 automated tests. I also build websites for individuals, businesses and research labs from

two templates of my own, in one or two languages.
rsilveirafh.github.io/modelos-de-site

- **Carona Solidária.** Ride-sharing platform for voters with reduced mobility in the 2026 runoff election, in three state capitals.
- **rr-quant.** Personal quant dashboard that runs itself every trading session (Python, Vercel): a pre-market snapshot of 46 assets, regime and correlation reads and a logistic-regression scoreboard for the Ibovespa.
github.com/rsilveirafh/rr-quant

## Education

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| **Postdoc**, Conservation, UFRN / PPBio | 2022–2023 |
| **Postdoc**, Plant ecology, Mato Grosso State University (UNEMAT) | 2018–2019 |
| **PhD in Zoology**, Federal University of Paraíba (UFPB) | 2017 |
| **MSc in Zoology**, Federal University of Paraíba (UFPB) | 2012 |
| **BSc in Biological Sciences (teaching)**, State University of Feira de Santana (UEFS) | 2009 |